

CAPSTONE PROJECT

OLAOLUWA LIVESTOCK FARM

Business Profile

Olaoluwa Livestock Farm is a small-scale farm business that deals with the rearing and sales of livestock animals. Its revenues come from direct sales, local markets, or supplying restaurants and processors.

Its digital landscape involves a lot of customer's data and a lot of modern devices used in catering for the farm animals, which includes;

- a. Customer's E-mail
- b. Customer's Address
- c. Payments System
- d. Livestock Feeding System
- e. Heating and cooling System
- f. Spreadsheet of farm invoices
- g. Employee's Record
- h. Database

Key assets to be identified

The key assets to be identified are what are paramount and important to the business that shouldn't be opened to vulnerabilities so that attackers who get in into the farm business system. This includes;

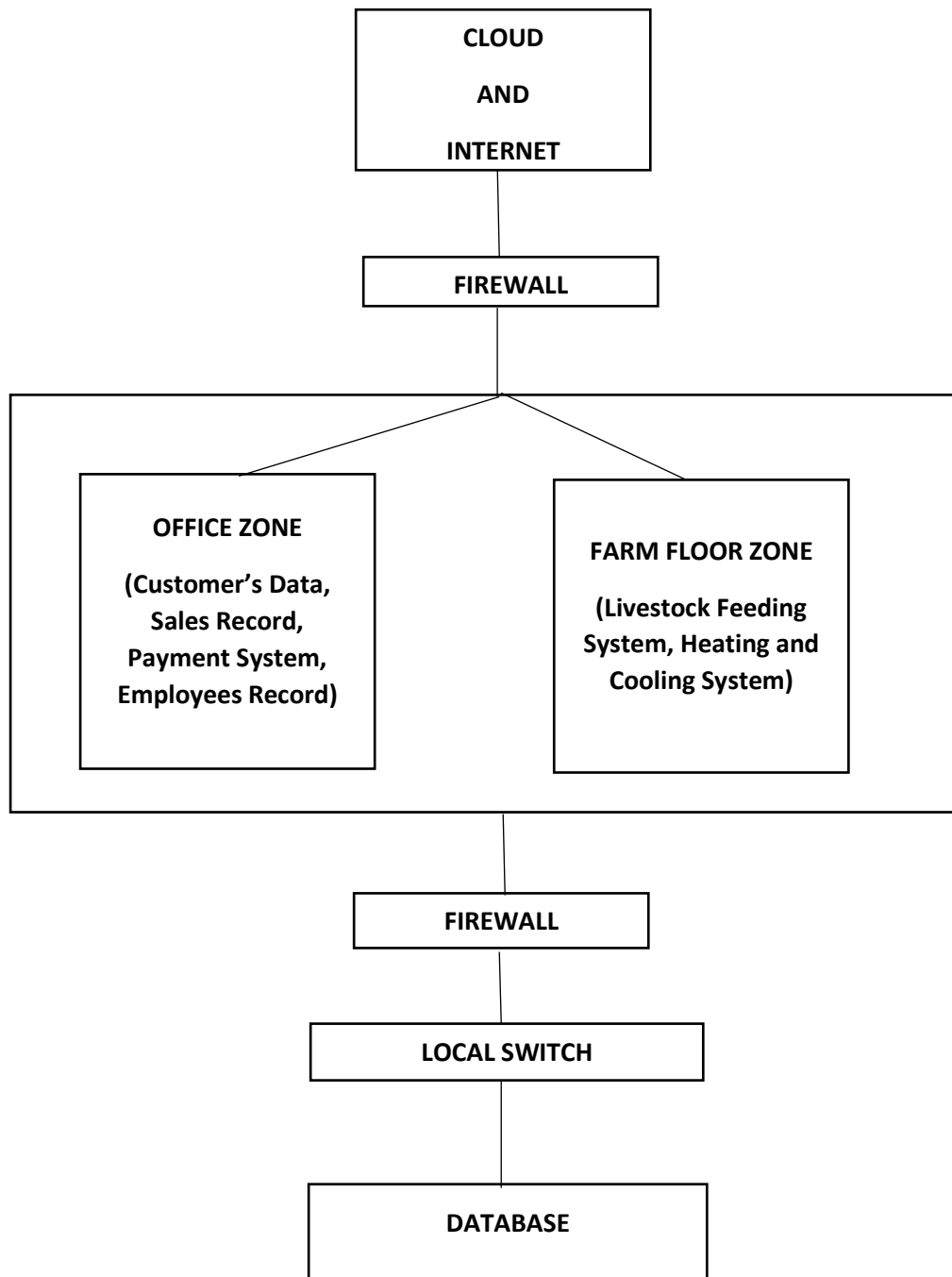
- a. **Customer's data:** Customer's data or records are one the key assets to be protected by the business because there has been confidentiality which made the customers interact with business. Confidentiality is a key factor in cybersecurity and it must hold for the business to be trusted by the customers.
- b. **Payment System:** The business payment is one of the most important key assets that must be protected. As we all know, one of the main reason an attacker attacks a system is for the attacker to be able to extort funds from the system owner or user so the payment system is a key assets that must be protected.
- c. **Database:** This where a lot of records are stored, almost every data in the system ends up in the database, it is a key assets the attackers are always after which must not be open to vulnerabilities.
- d. **Livestock Records:** This holds account for the livestock feeds invoice, the numbers of livestock animals we have on the farm, the sales record of the livestock animals. This is a key assets that must be protected cause it can be manipulated so as to extort the business for funds also.
- e. **Employee's Record:** Protecting the data of your employee is important so as not to expose them to social engineering by attackers

Realistic Threats the business is expose to.

- a. Impersonation

- b. Phishing attacks
- c. Social Engineering
- d. Malware Attacks

Recommend Security Architecture



Legal and Privacy Obligations

- a. **Registration:** Register your business with CAC and get necessary local permits

- b. **Tax:** register with FIRS, file returns, keep records for 6 years
- c. **Customer data:** Nigeria's NDPR requires you collect only what's needed, store it securely, and have a basic privacy policy
- d. **Payments:** use legitimate channels, keep transaction records
- e. **Staff:** written agreements, fair wages, safe conditions

How to prepare for emerging threats

- a. AI powered phishing targeting small businesses
- b. IoT vulnerabilities in feeding/heating systems
- c. Ransomware targeting farm management software

An incident response plan

- a. Identify — detect the attack
- b. Contain — isolate affected systems
- c. Eradicate — remove the threat
- d. Recover — restore systems from backup
- e. Review — document what happened and improve